

## ■ 시나리오 별 백업 복구(User Managed) - Archive log mode

### KEY WORDS

- <시나리오 5> 백업 받지 않은 테이블스페이스에 데이터파일 손상 되었을 경우
- <시나리오 6> 백업 받지 않은 TS 에 DFs 손상시 기존위치가 아닌 새로운 위치로 복구
- <시나리오 7> 특정한 데이터파일 손상으로 기존위치가 아닌 새로운 위치로 복구.
- <시나리오 8> TS 에 속한 여러 DFS 중에 특정한 file 이 손상되었을 경우 복구.

### <<시나리오 4>> 백업 받지 않은 테이블스페이스에 데이터파일 손상 되었을 경우

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
OPEN
```

```
SYS@orcl> select file#, name, checkpoint_change#, status from v$datafile
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS |
|-------|--------------------------------------------|--------------------|--------|
| 1     | /u01/app/oracle/oradata/orcl/system01.dbf  | 3386313            | SYSTEM |
| 2     | /u01/app/oracle/oradata/orcl/undotbs01.dbf | 3386313            | ONLINE |
| 3     | /u01/app/oracle/oradata/orcl/sysaux01.dbf  | 3386313            | ONLINE |
| 7     | /u01/app/oracle/oradata/orcl/users01.dbf   | 3386313            | ONLINE |

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# |
|-------|----------|--------------------------------------------|--------------------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2463719            |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2463719            |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2463719            |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2463719            |

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```

GROUP# SEQUENCE#
-----
MEMBER
-----
MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----
1          37
/u01/app/oracle/oradata/ORCL/redo01.log
10 INACTIVE        YES          2463466      2463469

2          39
/u01/app/oracle/oradata/ORCL/redo02.log
10 CURRENT         NO           2463472     1.8447E+19
```

```

GROUP# SEQUENCE#
-----
MEMBER
-----
MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----
3          38
/u01/app/oracle/oradata/ORCL/redo03.log
10 INACTIVE        YES          2463469      2463472
```

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#
```

```

FILE# NAME                                CHECKPOINT_CHANGE# STATUS
CHANGE# TIME
-----
1 /u01/app/oracle/oradata/ORCL/system01.dbf 2463719 NOT ACTIVE      2356080 2025-05-
```

10 19:36:51.70611

3 /u01/app/oracle/oradata/ORCL/sysaux01.dbf 2463719 NOT ACTIVE 2356080 2025-05-10 19:36:51.70611

4 /u01/app/oracle/oradata/ORCL/undotbs01.dbf 2463719 NOT ACTIVE 2356080 2025-05-10 19:36:51.70611

7 /u01/app/oracle/oradata/ORCL/users01.dbf 2463719 NOT ACTIVE 2356080 2025-05-10 19:36:51.70611

-- 온라인 백업본의 SCN 번호와 redo log의 SCN번호가 겹차이로 인한 변화가 많다면 온라인 백업.

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2463719            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2463719            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2463719            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2463719            | ONLINE |

SYS@orcl> create tablespace tbs datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' size 10m;  
Tablespace created.

SYS@orcl> create table hr.emp tablespace tbs as select \* from hr.employees;  
Table created.

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2463719            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2463719            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2463719            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2463719            | ONLINE |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2467504            | ONLINE |

```
SYS@orcl> update hr.emp set salary=2222 where employee_id=103;
1 row updated.
```

```
SYS@orcl> select salary from hr.emp where employee_id=103;
```

```

SALARY
-----
      2222

```

```
SYS@orcl> commit;
```

```
SYS@orcl> select f.file_name from dba_extents e, dba_data_files f
  2  where e.file_id = f.file_id and e.segment_name = 'EMP' and e.owner = 'HR';
```

```

FILE_NAME
-----
/u01/app/oracle/oradata/ORCL/tbs01.dbf

```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

| GROUP#  | SEQUENCE#  | MEMBER                                  | MB           | STATUS   |
|---------|------------|-----------------------------------------|--------------|----------|
|         | ARC        | FIRST_CHANGE#                           | NEXT_CHANGE# |          |
| 1       | 37         | /u01/app/oracle/oradata/ORCL/redo01.log | 10           | INACTIVE |
| 2463466 | 2463469    |                                         |              | YES      |
| 2       | 39         | /u01/app/oracle/oradata/ORCL/redo02.log | 10           | CURRENT  |
| 2463472 | 1.8447E+19 |                                         |              | NO       |
| 3       | 38         | /u01/app/oracle/oradata/ORCL/redo03.log | 10           | INACTIVE |
| 2463469 | 2463472    |                                         |              | YES      |

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

| GROUP#  | SEQUENCE#  | MEMBER                                  | MB           | STATUS   |
|---------|------------|-----------------------------------------|--------------|----------|
|         | ARC        | FIRST_CHANGE#                           | NEXT_CHANGE# |          |
| 1       | 43         | /u01/app/oracle/oradata/ORCL/redo01.log | 10           | CURRENT  |
| 2468983 | 1.8447E+19 |                                         |              | NO       |
| 2       | 42         | /u01/app/oracle/oradata/ORCL/redo02.log | 10           | ACTIVE   |
| 2468912 | 2468983    |                                         |              | YES      |
| 3       | 41         | /u01/app/oracle/oradata/ORCL/redo03.log | 10           | INACTIVE |
| 2468908 | 2468912    |                                         |              | YES      |

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

| SEQUENCE# | NAME                                        | FIRST_CHANGE# | NEXT_CHANGE# |
|-----------|---------------------------------------------|---------------|--------------|
| 30        | /home/oracle/arch1/arch_1_30_1200525254.arc | 2347853       | 2354351      |
| 30        | /home/oracle/arch2/arch_1_30_1200525254.arc | 2347853       | 2354351      |
| 31        | /home/oracle/arch1/arch_1_31_1200525254.arc | 2354351       | 2356143      |
| 31        | /home/oracle/arch2/arch_1_31_1200525254.arc | 2354351       | 2356143      |
| 32        | /home/oracle/arch1/arch_1_32_1200525254.arc | 2356143       | 2456244      |
| 32        | /home/oracle/arch2/arch_1_32_1200525254.arc | 2356143       | 2456244      |
| 33        | /home/oracle/arch1/arch_1_33_1200525254.arc | 2456244       | 2457117      |
| 33        | /home/oracle/arch2/arch_1_33_1200525254.arc | 2456244       | 2457117      |
| 34        | /home/oracle/arch1/arch_1_34_1200525254.arc | 2457117       | 2459505      |
| 34        | /home/oracle/arch2/arch_1_34_1200525254.arc | 2457117       | 2459505      |
| 35        | /home/oracle/arch1/arch_1_35_1200525254.arc | 2459505       | 2463463      |
| 35        | /home/oracle/arch2/arch_1_35_1200525254.arc | 2459505       | 2463463      |
| 36        | /home/oracle/arch1/arch_1_36_1200525254.arc | 2463463       | 2463466      |
| 36        | /home/oracle/arch2/arch_1_36_1200525254.arc | 2463463       | 2463466      |
| 37        | /home/oracle/arch1/arch_1_37_1200525254.arc | 2463466       | 2463469      |
| 37        | /home/oracle/arch2/arch_1_37_1200525254.arc | 2463466       | 2463469      |
| 38        | /home/oracle/arch1/arch_1_38_1200525254.arc | 2463469       | 2463472      |

|                                                |         |         |
|------------------------------------------------|---------|---------|
| 38 /home/oracle/arch2/arch_1_38_1200525254.arc | 2463469 | 2463472 |
| 39 /home/oracle/arch1/arch_1_39_1200525254.arc | 2463472 | 2468896 |
| 39 /home/oracle/arch2/arch_1_39_1200525254.arc | 2463472 | 2468896 |
| 40 /home/oracle/arch1/arch_1_40_1200525254.arc | 2468896 | 2468908 |
| 40 /home/oracle/arch2/arch_1_40_1200525254.arc | 2468896 | 2468908 |
| 41 /home/oracle/arch1/arch_1_41_1200525254.arc | 2468908 | 2468912 |
| 41 /home/oracle/arch2/arch_1_41_1200525254.arc | 2468908 | 2468912 |
| 42 /home/oracle/arch1/arch_1_42_1200525254.arc | 2468912 | 2468983 |
| 42 /home/oracle/arch2/arch_1_42_1200525254.arc | 2468912 | 2468983 |

32 rows selected.

### ■ 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/tbs01.dbf: No such file or directory
```

- 장애로 인해 데이터베이스가 비정상종료

```
SYS@orcl> shut immediate
ORA-03113: end-of-file on communication channel
Process ID: 17537
Session ID: 33 Serial number: 29608
```

```
-- Alert log file
2025-05-10T22:56:05.628136+09:00
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_lgwr_17496.trc:
ORA-63999: data file suffered media failure
ORA-01116: error in opening database file 5
ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/tbs01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

```
SYS@orcl> conn / as sysdba
Connected to an idle instance.
```

```
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               771751936 bytes
Database Buffers           352321536 bytes
Redo Buffers                7880704 bytes
```

```
Database mounted.
```

```
ORA-01157: cannot identify/lock data file 5 - see DBWR trace file
```

```
ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/tbs01.dbf'
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/tbs01.dbf: No such file or directory
```

```
SYS@orcl> select * from v$recover_file;
```

| FILE# | ONLINE | ONLINE_ ERROR | CHANGE#        |
|-------|--------|---------------|----------------|
| TIME  | CON_ID |               |                |
| ----- |        |               |                |
| 5     | ONLINE | ONLINE        | FILE NOT FOUND |
| 0     |        | 0             |                |

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS         | CHANGE# | TIME                      |
|-------|--------------------------------------------|--------------------|----------------|---------|---------------------------|
| ----- |                                            |                    |                |         |                           |
| 1     | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2468983            | NOT ACTIVE     | 2356080 | 2025-05-10 19:36:51.70611 |
| 3     | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2468983            | NOT ACTIVE     | 2356080 | 2025-05-10 19:36:51.70611 |
| 4     | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2468983            | NOT ACTIVE     | 2356080 | 2025-05-10 19:36:51 70611 |
| 5     | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2468983            | FILE NOT FOUND |         | 0                         |
| 7     | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2468983            | NOT ACTIVE     | 2356080 | 2025-05-10 19:36:51.7061  |

## ■ Recovery

### 1. 복구 수행해야할 데이터파일 offline

```
SYS@orcl> alter database datafile 5 offline;
Database altered.
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS  |
|-------|----------|--------------------------------------------|--------------------|---------|
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2468983            | SYSTEM  |
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2468983            | ONLINE  |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2468983            | ONLINE  |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2468983            | ONLINE  |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2468983            | RECOVER |

### 2. 데이터베이스 open

```
SYS@orcl> alter database open;
Database altered.
```

```
SYS@orcl> select count(*) from hr.employees;
```

```
COUNT(*)
-----
107
```

### 3. 백업받지 않은 테이블스페이스에 속한 파일을 재생성

```
SYS@orcl> select count(*) from hr.emp;
select count(*) from hr.emp
*
```

ERROR at line 1:

ORA-00376: file 5 cannot be read at this time

ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/tbs01.dbf'

```
SYS@orcl> alter database create datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf';
Database altered.
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs01.dbf
```



/u01/app/oracle/oradata/ORCL/tbs01.dbf

#### 4. RECOVER 명령어(어느 것이든 하나 수행)재생성 이후 변경된 리두정보 적용

```
SYS@orcl> alter database recover datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf';
```

```
SYS@orcl> recover datafile 5
```

```
SYS@orcl> recover datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf'
```

```
-----
```

```
SYS@orcl> recover datafile 5;
```

```
ORA-00279: change 2467503 generated at 05/10/2025 22:36:28 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_39_1200525254.arc
```

```
ORA-00280: change 2467503 for thread 1 is in sequence #39
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
```

```
auto
```

```
ORA-00279: change 2468896 generated at 05/10/2025 22:42:15 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_40_1200525254.arc
```

```
ORA-00280: change 2468896 for thread 1 is in sequence #40
```

```
ORA-00279: change 2468908 generated at 05/10/2025 22:42:35 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_41_1200525254.arc
```

```
ORA-00280: change 2468908 for thread 1 is in sequence #41
```

```
Log applied.
```

```
Media recovery complete.
```

```
-- Alert log file 확인
```

#### 5. 복구 완료된 데이터 파일 online

```
SYS@orcl> alter database datafile 5 online;
```

```
Database altered.
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2569465            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2569465            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2569465            | ONLINE |

|         |                                          |                |
|---------|------------------------------------------|----------------|
| 7 USERS | /u01/app/oracle/oradata/ORCL/users01.dbf | 2569465 ONLINE |
| 5 TBS   | /u01/app/oracle/oradata/ORCL/tbs01.dbf   | 2570402 ONLINE |

SSYS@orcl> select salary from hr.emp where employee\_id=103;

SALARY

-----

2222

-- 백업 수행

SYS@orcl> !rm /home/oracle/arch1/\*

SYS@orcl> !rm /home/oracle/arch2/\*

SYS@orcl> !mkdir -p /home/oracle/backup/open\_bkp/20250512

SYS@orcl> alter database begin backup;

Database altered.

SYS@orcl> !cp -v /u01/oracle/app/oradata/ORCL/\* /home/oracle/backup/open\_bkp/20250512/

cp: cannot stat '/u01/oracle/app/oradata/ORCL/\*': No such file or directory

SYS@orcl> !cp -v /u01/app/oracle/oradata/ORCL/\* /home/oracle/backup/open\_bkp/20250512/

'/u01/app/oracle/oradata/ORCL/control01.ctl' -> '/home/oracle/backup/open\_bkp/20250512/control01.ctl'

'/u01/app/oracle/oradata/ORCL/control02.ctl' -> '/home/oracle/backup/open\_bkp/20250512/control02.ctl'

'/u01/app/oracle/oradata/ORCL/initiorcl.ora' -> '/home/oracle/backup/open\_bkp/20250512/initiorcl.ora'

'/u01/app/oracle/oradata/ORCL/redo01.log' -> '/home/oracle/backup/open\_bkp/20250512/redo01.log'

'/u01/app/oracle/oradata/ORCL/redo02.log' -> '/home/oracle/backup/open\_bkp/20250512/redo02.log'

'/u01/app/oracle/oradata/ORCL/redo03.log' -> '/home/oracle/backup/open\_bkp/20250512/redo03.log'

'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> '/home/oracle/backup/open\_bkp/20250512/sysaux01.dbf'

'/u01/app/oracle/oradata/ORCL/system01.dbf' -> '/home/oracle/backup/open\_bkp/20250512/system01.dbf'

'/u01/app/oracle/oradata/ORCL/tbs01.dbf' -> '/home/oracle/backup/open\_bkp/20250512/tbs01.dbf'

'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> '/home/oracle/backup/open\_bkp/20250512/temp01.dbf'

'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> '/home/oracle/backup/open\_bkp/20250512/undotbs01.dbf'

'/u01/app/oracle/oradata/ORCL/users01.dbf' -> '/home/oracle/backup/open\_bkp/20250512/users01.dbf'

SYS@orcl> alter database end backup;

Database altered.

## &lt;&lt;시나리오 6&gt;&gt; 백업 받지 않은 TS에 DFs 손상시 기존위치가 아닌 새로운 위치로 복구

```
SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
-rw-r-----. 1 oracle dba 10493952 May 10 23:12 /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS     | CHANGE# | TIME                      |
|-------|--------------------------------------------|--------------------|------------|---------|---------------------------|
| 1     | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2570617            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 3     | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2570617            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 4     | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2570617            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 5     | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2570617            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 7     | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2570617            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.3445  |

# 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/bs01.dbf
```

```
/u01/app/oracle/oradata/orcl/data_tbs01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/bs01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/orcl/data_tbs01.dbf: No such file or directory
```

```
SYS@orcl> conn / as sysdba
```

```
Connected to an idle instance.
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size 8895728 bytes
```

```
Variable Size 771751936 bytes
```

```
Database Buffers 352321536 bytes
```

Redo Buffers 7880704 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 5 - see DBWR trace file

ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/tbs01.dbf'

-Alert log file

ORA-1157 signalled during: ALTER DATABASE OPEN...

2025-05-10T23:22:11.292128+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl\_mz00\_25403.trc:

ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/tbs01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/tbs01.dbf'

ORA-27037: unable to obtain file status

Linux-x86\_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> select \* from v\$recover\_file;

| FILE#        | ONLINE | ONLINE_ | ERROR          |
|--------------|--------|---------|----------------|
| CHANGE# TIME | CON_ID |         |                |
| -----        | -----  | -----   | -----          |
| 5            | ONLINE | ONLINE  | FILE NOT FOUND |
| 0            | 0      |         |                |

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| ----- | -----    | -----                                      | -----              | -----  |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2570617            | SYSTEM |
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2570617            | ONLINE |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2570617            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2570617            | ONLINE |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2570617            | ONLINE |

## ■ Recovery

1. 복구 수행해야 할 데이터파일 offline

SYS@orcl> alter database datafile 5 offline;

Database altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE#              | TBS_NAME | FILE_NAME                                   |
|--------------------|----------|---------------------------------------------|
| CHECKPOINT_CHANGE# | STATUS   |                                             |
| -----              |          |                                             |
| 1                  | SYSTEM   | /u01/app/oracle/oradata/orcl/system01.dbf   |
| 3491965            | SYSTEM   |                                             |
| 3                  | SYSAUX   | /u01/app/oracle/oradata/orcl/sysaux01.dbf   |
| 3491965            | ONLINE   |                                             |
| 2                  | UNDOTBS1 | /u01/app/oracle/oradata/orcl/undotbs01.dbf  |
| 3491965            | ONLINE   |                                             |
| 7                  | USERS    | /u01/app/oracle/oradata/orcl/users01.dbf    |
| 3491965            | ONLINE   |                                             |
| 5                  | DATA_TBS | /u01/app/oracle/oradata/orcl/data_tbs01.dbf |
| 3492977            | RECOVER  |                                             |

```
-- recover 확인
```

## 2. 데이터베이스 open

```
SYS@orcl> alter database open;
Database altered.
```

```
SYS@orcl> select count(*) from hr.emp;
select count(*) from hr.emp
*
```

```
ERROR at line 1:
ORA-00376: file 5 cannot be read at this time
ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/tbs01.dbf'
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/tbs01.dbf: No such file or directory
```

```
SYS@orcl> alter database create datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf' as '/home/oracle/tbs01.dbf';

Database altered.
```

```
SYS@orcl> !ls /home/oracle/tbs01.dbf
```

/home/oracle/tbs01.dbf

### 3. RECOVER 명령어(어느 것이든 하나 수행)재생성 이후 변경된 리두정보 적용

```
SYS@orcl>alter database recover datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf';
```

```
SYS@orcl>recover datafile 5
```

```
SYS@orcl>recover datafile '/u01/app/oracle/oradata/ORCL/tbs01.dbf'
```

-----

```
SYS@orcl> recover datafile 5;
```

```
ORA-00279: change 2467503 generated at 05/10/2025 22:36:28 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_39_1200525254.arc
```

```
ORA-00280: change 2467503 for thread 1 is in sequence #39
```

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

auto

```
ORA-00279: change 2468896 generated at 05/10/2025 22:42:15 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_40_1200525254.arc
```

```
ORA-00280: change 2468896 for thread 1 is in sequence #40
```

```
ORA-00279: change 2468908 generated at 05/10/2025 22:42:35 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_41_1200525254.arc
```

```
ORA-00280: change 2468908 for thread 1 is in sequence #41
```

```
ORA-00279: change 2468912 generated at 05/10/2025 22:42:39 needed for thread 1
```

```
ORA-00289: suggestion : /home/oracle/arch2/arch_1_42_1200525254.arc
```

```
ORA-00280: change 2468912 for thread 1 is in sequence #42
```

Log applied.

Media recovery complete.

-- Alert log file

2025-05-10T23:30:15.206612+09:00

ALTER DATABASE RECOVER CONTINUE DEFAULT

2025-05-10T23:30:15.206796+09:00

Media Recovery Log /home/oracle/arch2/arch\_1\_39\_1200525254.arc

ORA-279 signalled during: ALTER DATABASE RECOVER CONTINUE DEFAULT ...

ALTER DATABASE RECOVER CONTINUE DEFAULT

2025-05-10T23:30:15.224295+09:00

Media Recovery Log /home/oracle/arch2/arch\_1\_40\_1200525254.arc

ORA-279 signalled during: ALTER DATABASE RECOVER CONTINUE DEFAULT ...

ALTER DATABASE RECOVER CONTINUE DEFAULT

2025-05-10T23:30:15.236288+09:00

Media Recovery Log /home/oracle/arch2/arch\_1\_41\_1200525254.arc

ORA-279 signalled during: ALTER DATABASE RECOVER CONTINUE DEFAULT ...

ALTER DATABASE RECOVER CONTINUE DEFAULT

2025-05-10T23:30:15.245775+09:00

Media Recovery Log /home/oracle/arch2/arch\_1\_42\_1200525254.arc

2025-05-10T23:30:15.253291+09:00

Recovery of Online Redo Log: Thread 1 Group 1 Seq 43 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo01.log

2025-05-10T23:30:15.255751+09:00

Recovery of Online Redo Log: Thread 1 Group 3 Seq 44 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo03.log

2025-05-10T23:30:15.270100+09:00

Recovery of Online Redo Log: Thread 1 Group 2 Seq 45 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo02.log

2025-05-10T23:30:15.275475+09:00

Media Recovery Complete (orcl)

Completed: ALTER DATABASE RECOVER CONTINUE DEFAULT

#### 4. 복구 완료된 데이터 파일 online

SSYS@orcl> alter database datafile 5 online;

Database altered.

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2671899            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2671899            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2671899            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2671899            | ONLINE |
| 5     | TBS      | /home/oracle/tbs01.dbf                     | 2672874            | ONLINE |

## ■ 특정한 테이블스페이스에 대해서 백업

```
# alter tablespace 테이블명 begin backup;
```

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS     | CHANGE# | TIME                      |
|-------|--------------------------------------------|--------------------|------------|---------|---------------------------|
| 1     | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2773392            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 3     | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2773392            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 4     | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2773392            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 5     | /home/oracle/tbs01.dbf                     | 2774349            | NOT ACTIVE | 0       |                           |
| 7     | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2773392            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.3445  |

```
SYS@orcl> alter tablespace tbs begin backup;
```

Tablespace altered.

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS     | CHANGE# | TIME                      |
|-------|--------------------------------------------|--------------------|------------|---------|---------------------------|
| 1     | /u01/app/oracle/oradata/orcl/system01.dbf  | 3594588            | NOT ACTIVE | 3279657 | 2025-02-19 14:35:39.52539 |
| 2     | /u01/app/oracle/oradata/orcl/undotbs01.dbf | 3594588            | NOT ACTIVE | 3279657 | 2025-02-19 14:35:39.52539 |
| 3     | /u01/app/oracle/oradata/orcl/sysaux01.dbf  | 3594588            | NOT ACTIVE | 3279657 | 2025-02-19 14:35:39.52539 |
| 5     | /home/oracle/data_tbs01.dbf                | 3596325            | ACTIVE     | 3596325 | 2025-02-20 11:59:41.43181 |
| 7     | /u01/app/oracle/oradata/orcl/users01.dbf   | 3594588            | NOT ACTIVE | 3279657 | 2025-02-19 14:35:39.52539 |



```
SYS@orcl> ! cp -v /home/oracle/tbs01.dbf /home/oracle/backup/open_bkp/20250512/  
'/home/oracle/tbs01.dbf' -> '/home/oracle/backup/open_bkp/20250512/tbs01.dbf'
```

```
SYS@orcl> alter tablespace tbs end backup;
```

Tablespace altered.

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd  
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file# and a.file#=5;
```

| FILE# | NAME                   | CHECKPOINT_CHANGE# | STATUS     | CHANGE# | TIME                      |
|-------|------------------------|--------------------|------------|---------|---------------------------|
| 5     | /home/oracle/tbs01.dbf | 2774385            | NOT ACTIVE | 2774385 | 2025-05-10 23:51:22.85882 |

<<시나리오 7>> 특정한 데이터파일 손상으로 기존위치가 아닌 새로운 위치로 복구.

```
SYS@orcl> !ls -l /home/oracle/tbs01.dbf
```

```
-rw-r-----. 1 oracle dba 10493952 May 10 23:31 /home/oracle/tbs01.dbf
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2773392            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2773392            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2773392            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2773392            | ONLINE |
| 5     | TBS      | /home/oracle/tbs01.dbf                     | 2774385            | ONLINE |

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS     | CHANGE# | TIME                       |
|-------|--------------------------------------------|--------------------|------------|---------|----------------------------|
| 1     | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2773392            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45. 83445 |
| 3     | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2773392            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45. 83445 |
| 4     | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2773392            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45. 83445 |
| 5     | /home/oracle/tbs01.dbf                     | 2774385            | NOT ACTIVE | 2774385 | 2025-05-10 23:51:22. 85882 |

# 장애발생

```
SYS@orcl> !ls /home/oracle/tbs01.dbf
/home/oracle/bs01.dbf
```

```
SYS@orcl> !rm /home/oracle/tbs01.dbf
```

```
SYS@orcl> !ls /home/oracle/data_tbs01.dbf
```

```
ls: cannot access /home/oracle/data_tbs01.dbf: No such file or directory
```

```
SYS@orcl> shutdown immediate
ORA-03113: end-of-file on communication channel
Process ID: 26851
Session ID: 1 Serial number: 29231
```

```
-- Alert log file
22025-05-10T23:59:06.440154+09:00
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_lgwr_26811.trc:
ORA-63999: data file suffered media failure
ORA-01116: error in opening database file 5
ORA-01110: data file 5: '/home/oracle/tbs01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

```
SYS@orcl> conn /as sysdba
Connected to an idle instance.
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               771751936 bytes
Database Buffers           352321536 bytes
Redo Buffers                7880704 bytes
Database mounted.
ORA-01157: cannot identify/lock data file 5 - see DBWR trace file
ORA-01110: data file 5: '/home/oracle/tbs01.dbf'
```

```
SYS@orcl> select * from v$recover_file;
```

| FILE# | ONLINE | ONLINE_ ERROR | CHANGE#        |
|-------|--------|---------------|----------------|
| TIME  | CON_ID |               |                |
| ----- |        |               |                |
| 5     | ONLINE | ONLINE        | FILE NOT FOUND |
| 0     |        | 0             |                |

## ■ Recovery

### 1. 문제가 발생한 데이터파일 offline

```
SSYS@orcl> alter database datafile 5 offline;
```

Database altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS  |
|-------|----------|--------------------------------------------|--------------------|---------|
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2671899            | SYSTEM  |
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2671899            | ONLINE  |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2671899            | ONLINE  |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2671899            | ONLINE  |
| 5     | TBS      | /home/oracle/tbs01.dbf                     | 2672874            | RECOVER |

### 2. 데이터베이스 open

```
SYS@orcl> alter database open;
```

Database altered.

### 3. 최근 백업본을 찾아 새로운 위치로 restore

```
SYS@orcl> SYS@orcl> !ls /home/oracle/backup/open_bkp/20250512/tbs01.dbf
/home/oracle/backup/open_bkp/20250512/tbs01.dbf
```

```
SYS@orcl> ! cp /home/oracle/backup/open_bkp/20250512/tbs01.dbf /u01/app/oracle/oradata/ORCL/
```

```
SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/
control01.ctl  initorcl.ora  redo02.log  sysaux01.dbf  tbs01.dbf  undotbs01.dbf
control02.ctl  redo01.log    redo03.log  system01.dbf  temp01.dbf  users01.dbf
```

### 4. control file에 기존위치가 아닌 새로운 위치로 변경

```
alter database rename file '기존위치' to '새로운위치';
```

```
SYS@orcl> alter database rename file '/home/oracle/tbs01.dbf' to '/u01/app/oracle/oradata/ORCL/tbs01.dbf';
```

Database altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS  |
|-------|----------|--------------------------------------------|--------------------|---------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2874946            | ONLINE  |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2874946            | SYSTEM  |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2874946            | ONLINE  |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2874946            | ONLINE  |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2774385            | RECOVER |

#### 5. 백업 이후 변경정보(리두) 적용

```
SYS@orcl> recover tablespace tbs;
Media recovery complete.
```

-- Alert log file

2025-05-11T00:04:56.817614+09:00

```
alter database rename file '/home/oracle/tbs01.dbf' to '/u01/app/oracle/oradata/ORCL/tbs01.dbf'
```

Completed: alter database rename file '/home/oracle/tbs01.dbf' to '/u01/app/oracle/oradata/ORCL/tbs01.dbf'

2025-05-11T00:05:35.932767+09:00

```
ALTER DATABASE RECOVER tablespace tbs
```

2025-05-11T00:05:35.933136+09:00

Media Recovery Start

2025-05-11T00:05:35.933360+09:00

Serial Media Recovery started

2025-05-11T00:05:35.969906+09:00

Recovery of Online Redo Log: Thread 1 Group 1 Seq 46 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo01.log

2025-05-11T00:05:35.979440+09:00

Recovery of Online Redo Log: Thread 1 Group 3 Seq 47 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo03.log

2025-05-11T00:05:35.985054+09:00

Media Recovery Complete (orcl)

Completed: ALTER DATABASE RECOVER tablespace tbs

6. 복구 완료된 테이블스페이스 online

```
SYS@orcl> alter tablespace tbs online;
```

Tablespace altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,  
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2874946            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2874946            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2874946            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2874946            | ONLINE |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2875838            | ONLINE |

```
SYS@orcl> select salary from hr.emp where employee_id=103;
```

| SALARY |
|--------|
| 2222   |

## &lt;&lt;시나리오 8&gt;&gt; TS에 속한 여러 DFs 중에 특정한 file이 손상되었을 경우 복구

```
SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
-rw-r-----. 1 oracle dba 10493952 May 11 00:06 /u01/app/oracle/oradata/ORCL/tbs01.dbf
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2874946            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2874946            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2874946            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2874946            | ONLINE |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2875838            | ONLINE |

# 테이블스페이스 데이터파일 추가

```
SYS@orcl> alter tablespace tbs add datafile '/u01/app/oracle/oradata/ORCL/tbs02.dbf' size 10m;
```

Tablespace altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2874946            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2874946            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2874946            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2874946            | ONLINE |
| 2     | TBS      | /u01/app/oracle/oradata/ORCL/tbs02.dbf     | 2876224            | ONLINE |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2875838            | ONLINE |

```
SYS@orcl> create table hr.emp1 tablespace tbs as select * from dba_objects;
```

Table created.

```
SYS@orcl> create table hr.emp1 tablespace tbs as select * from dba_objects;
```

Table created.

```
SYS@orcl> select f.file_name from dba_extents e, dba_data_files f
where e.file_id = f.file_id and e.segment_name = 'EMP1' and e.owner = 'HR';  2
```

FILE\_NAME

```
-----
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
```

FILE\_NAME

```
-----
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs01.dbf
/u01/app/oracle/oradata/ORCL/tbs02.dbf
```

FILE\_NAME

```
-----
/u01/app/oracle/oradata/ORCL/tbs02.dbf
/u01/app/oracle/oradata/ORCL/tbs02.dbf
/u01/app/oracle/oradata/ORCL/tbs02.dbf
/u01/app/oracle/oradata/ORCL/tbs02.dbf
/u01/app/oracle/oradata/ORCL/tbs02.dbf
```



27 rows selected.

```
SYS@orcl> select f.tablespace_name, f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id
and e.segment_name = 'EMP1' and e.owner = 'HR' group by f.tablespace_name, f.file_name;
```

| TABLESPACE_NAME | FILE_NAME                              |
|-----------------|----------------------------------------|
| TBS             | /u01/app/oracle/oradata/ORCL/tbs02.dbf |
| TBS             | /u01/app/oracle/oradata/ORCL/tbs01.dbf |

```
SYS@orcl> select f.tablespace_name, f.file_name, count(*) from dba_extents e, dba_data_files f where e.file_id =
f.file_id and e.segment_name = 'EMP1' and e.owner = 'HR' group by f.tablespace_name, f.file_name;
```

| TABLESPACE_NAME | FILE_NAME                              | COUNT(*) |
|-----------------|----------------------------------------|----------|
| TBS             | /u01/app/oracle/oradata/ORCL/tbs02.dbf | 6        |
| TBS             | /u01/app/oracle/oradata/ORCL/tbs01.dbf | 21       |

```
SYS@orcl> select bytes/1024/1024 mb
from dba_segments
where segment_name = 'EMP1'
and owner = 'HR'; 2 3 4
```

| MB |
|----|
| 12 |

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE# | NAME                                      | CHECKPOINT_CHANGE# | STATUS     | CHANGE# | TIME       |
|-------|-------------------------------------------|--------------------|------------|---------|------------|
| 1     | /u01/app/oracle/oradata/ORCL/system01.dbf | 2874946            | NOT ACTIVE | 2570617 | 2025-05-10 |

23:10:45.

83445

|                                             |                    |                    |
|---------------------------------------------|--------------------|--------------------|
| 2 /u01/app/oracle/oradata/ORCL/tbs02.dbf    | 2876224 NOT ACTIVE | 0                  |
| 3 /u01/app/oracle/oradata/ORCL/sysaux01.dbf | 2874946 NOT ACTIVE | 2570617 2025-05-10 |

23:10:45.

83445

|                                              |                    |         |
|----------------------------------------------|--------------------|---------|
| 4 /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2874946 NOT ACTIVE | 2570617 |
|----------------------------------------------|--------------------|---------|

2025-05-10

23:10:45.

| FILE#   | NAME | CHECKPOINT_CHANGE# | STATUS |
|---------|------|--------------------|--------|
| CHANGE# | TIME |                    |        |

83445

|                                          |                    |         |
|------------------------------------------|--------------------|---------|
| 5 /u01/app/oracle/oradata/ORCL/tbs01.dbf | 2875838 NOT ACTIVE | 2774385 |
|------------------------------------------|--------------------|---------|

2025-05-10

23:51:22.

85882

|                                            |                    |                    |
|--------------------------------------------|--------------------|--------------------|
| 7 /u01/app/oracle/oradata/ORCL/users01.dbf | 2874946 NOT ACTIVE | 2570617 2025-05-10 |
|--------------------------------------------|--------------------|--------------------|

23:10:45.

83445

6 rows selected.

# 특정한 테이블스페이스에 속한 데이터파일 백업

```
SYS@orcl> alter tablespace tbs begin backup;
```

Tablespace altered.

```
SYS@orcl> ! cp -v /u01/app/oracle/oradata/ORCL/tbs02.dbf /home/oracle/backup/open_bkp/20250512
```

```
SYS@orcl> alter tablespace tbs end backup;
```

Tablespace altered.

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS     | CHANGE# | TIME                      |
|-------|--------------------------------------------|--------------------|------------|---------|---------------------------|
| 1     | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2878252            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 2     | /u01/app/oracle/oradata/ORCL/tbs02.dbf     | 2878367            | NOT ACTIVE | 2878367 | 2025-05-11 00:22:30.01350 |
| 3     | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2878252            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 4     | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2878252            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |
| 5     | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2878367            | NOT ACTIVE | 2878367 | 2025-05-11 00:22:30.01350 |
| 7     | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2878252            | NOT ACTIVE | 2570617 | 2025-05-10 23:10:45.83445 |

6 rows selected.

```
SYS@orcl> ! ls /home/oracle/backup/open_bkp/20250512
```

```
control01.ctl tbs01.dbf tbs02.dbf sysaux01.dbf system01.dbf temp01.dbf undotbs01.dbf users01.dbf
```

```
SSYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

| GROUP# | SEQUENCE#     | MEMBER                                  | MB | STATUS       |
|--------|---------------|-----------------------------------------|----|--------------|
| ARC    | FIRST_CHANGE# | NEXT_CHANGE#                            |    |              |
| 1      | 46            | /u01/app/oracle/oradata/ORCL/redo01.log | 10 | INACTIVE YES |

```

2773391      2874945
           2      48 /u01/app/oracle/oradata/ORCL/redo02.log          10 CURRENT          NO
2878252  1.8447E+19
           3      47 /u01/app/oracle/oradata/ORCL/redo03.log          10 INACTIVE          YES
2874945      2878252

```

SYS@orcl> select sequence#, name, FIRST\_CHANGE#, NEXT\_CHANGE# from v\$archived\_log;

| SEQUENCE# | NAME                                        | FIRST_CHANGE# | NEXT_CHANGE# |
|-----------|---------------------------------------------|---------------|--------------|
| 43        | /home/oracle/arch1/arch_1_43_1200525254.arc | 2468983       | 2569464      |
| 43        | /home/oracle/arch2/arch_1_43_1200525254.arc | 2468983       | 2569464      |
| 44        | /home/oracle/arch1/arch_1_44_1200525254.arc | 2569464       | 2671898      |
| 44        | /home/oracle/arch2/arch_1_44_1200525254.arc | 2569464       | 2671898      |
| 45        | /home/oracle/arch1/arch_1_45_1200525254.arc | 2671898       | 2773391      |
| 45        | /home/oracle/arch2/arch_1_45_1200525254.arc | 2671898       | 2773391      |
| 46        | /home/oracle/arch1/arch_1_46_1200525254.arc | 2773391       | 2874945      |
| 46        | /home/oracle/arch2/arch_1_46_1200525254.arc | 2773391       | 2874945      |
| 47        | /home/oracle/arch1/arch_1_47_1200525254.arc | 2874945       | 2878252      |
| 47        | /home/oracle/arch2/arch_1_47_1200525254.arc | 2874945       | 2878252      |

42 rows selected.

SYS@orcl> alter system switch logfile;

System altered.

SYS@orcl> /

System altered.

SYS@orcl> /

System altered.

SYS@orcl> create table hr.dept1 tablespace tbs as select \* from hr.departments;

Table created.

```
SYS@orcl> select f.file_name from dba_extents e, dba_data_files f
where e.file_id = f.file_id and e.segment_name = 'DEPT1' and e.owner = 'HR';    2
```

```
FILE_NAME
```

```
-----
/u01/app/oracle/oradata/ORCL/tbs02.dbf
```

# 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs02.dbf
/u01/app/oracle/oradata/ORCL/tbs02.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/tbs02.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs02.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/tbs02.dbf: No such file or directory
```

1. 테이블 스페이스에 속한 데이터파일을 offline으로 수행하되 문제가 없는 데이터파일은 checkpoint 발생하지 않음

```
SYS@orcl> alter tablespace tbs offline temporary;
```

Tablespace altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS  |
|-------|----------|--------------------------------------------|--------------------|---------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2878546            | ONLINE  |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2878546            | SYSTEM  |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2878546            | ONLINE  |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2878546            | ONLINE  |
| 2     | TBS      | /u01/app/oracle/oradata/ORCL/tbs02.dbf     | 2878921            | OFFLINE |
| 5     | TBS      | /u01/app/oracle/oradata/ORCL/tbs01.dbf     | 2878921            | OFFLINE |

6 rows selected.

2. 문제가 발생한 데이터파일에 대해서 최근 백업을 찾아 restore

```
SYS@orcl> ! ls /home/oracle/backup/open_bkp/20250512/tbs02.dbf
/home/oracle/backup/open_bkp/20250512/tbs02.dbf
```

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/20250512/tbs02.dbf /u01/app/oracle/oradata/ORCL/tbs02.dbf
'/home/oracle/backup/open_bkp/20250512/tbs02.dbf' -> '/u01/app/oracle/oradata/ORCL/tbs02.dbf'
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/tbs02.dbf
/u01/app/oracle/oradata/ORCL/tbs02.dbf
```

### 3. 백업이후에 변경정보(리두) 적용

```
SYS@orcl> recover tablespace tbs
ORA-00279: change 2878367 generated at 05/11/2025 00:22:30 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_48_1200525254.arc
ORA-00280: change 2878367 for thread 1 is in sequence #48
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto
ORA-00279: change 2878537 generated at 05/11/2025 00:27:52 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_49_1200525254.arc
ORA-00280: change 2878537 for thread 1 is in sequence #49
```

Log applied.  
Media recovery complete.

-- Alert log file

```
2025-05-11T00:37:32.166599+09:00
ALTER DATABASE RECOVER tablespace tbs
2025-05-11T00:37:32.166767+09:00
Media Recovery Start
2025-05-11T00:37:32.167352+09:00
Serial Media Recovery started
ORA-279 signalled during: ALTER DATABASE RECOVER tablespace tbs ...
2025-05-11T00:37:35.791164+09:00
ALTER DATABASE RECOVER CONTINUE DEFAULT
2025-05-11T00:37:35.796180+09:00
Media Recovery Log /home/oracle/arch2/arch_1_48_1200525254.arc
```

ORA-279 signalled during: ALTER DATABASE RECOVER CONTINUE DEFAULT ...

ALTER DATABASE RECOVER CONTINUE DEFAULT

2025-05-11T00:37:35.800738+09:00

Media Recovery Log /home/oracle/arch2/arch\_1\_49\_1200525254.arc

2025-05-11T00:37:35.807574+09:00

Recovery of Online Redo Log: Thread 1 Group 3 Seq 50 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo03.log

2025-05-11T00:37:35.807976+09:00

Recovery of Online Redo Log: Thread 1 Group 2 Seq 51 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo02.log

2025-05-11T00:37:35.818572+09:00

Recovery of Online Redo Log: Thread 1 Group 1 Seq 52 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo01.log

2025-05-11T00:37:35.835948+09:00

Media Recovery Complete (orcl)

Completed: ALTER DATABASE RECOVER CONTINUE DEFAULT

4. 복구 완료된 테이블스페이스 online

SYS@orcl> alter tablespace tbs online;

Tablespace altered.

SSYS@orcl> select count(\*) from hr.emp1;

COUNT(\*)

-----

72449

SYS@orcl> select count(\*) from hr.dept1;

COUNT(\*)

-----

27

SYS@orcl> drop tablespace tbs including contents and datafiles;

Tablespace dropped.

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a,

v\$tablespace b where a.ts# = b.ts#;

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2878546            | ONLINE |
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2878546            | SYSTEM |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2878546            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2878546            | ONLINE |

```
SYS@orcl> ! ls /home/oracle/backup/open_bkp/20250512/tbs*  
/home/oracle/backup/open_bkp/20250512/tbs01.dbf  
/home/oracle/backup/open_bkp/20250512/tbs02.dbf
```

```
SYS@orcl> ! rm /home/oracle/backup/open_bkp/20250512/tbs*
```